

A gap in the proof of Lemma 3.12 of
Cavenagh–Greenhill–Wanless,
The cycle structure of two rows in a random Latin square

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Abstract

Case 3 of the splitting procedure used to prove Lemma 3.12 of [1] (cross-switch with respect to $\{\omega(j), \omega(j')\}$, then backflip with respect to $\{j, j'\}$) does not behave as claimed: the cross-switch reverses an ω -arc containing j' (or j), after which $\{j, j'\}$ is at ω -distance 2. When $\min(\alpha, \beta) \geq 3$ the backflip therefore produces a square of the wrong type. When $\min(\alpha, \beta) = 2 < \max(\alpha, \beta)$ the type is right, but the resulting edge is in general not an edge of the joining multigraph, so the identity $G_S = G_J$ on which the double count rests fails; already for $n = 5$, the sharp case, the splitting multigraph has vertices of degree 24 on the $S(\mu, F)$ side where the proof asserts at most 18. The lower bound of Lemma 3.12 survives (and is in fact elementary); the upper bound $3/2$, hence the upper bound of Theorem 3.13 and the results of Section 4 that use it (listed below), is not established by the published argument for any split other than $(2, 2)$. A canonical repair of case 3 exists when α is even and $\alpha < 2\beta$ (or symmetrically); we do not see one when α, β are both odd. Exact data for $n \leq 11$ and sampled squares up to $n = 100$ indicate that the statement is true with much room to spare.

1 Setting and the procedure

We use the notation of [1], Section 3. L is a Latin square of order n containing a suitable partial square F ; ω is the permutation of the first m columns with $1 \circ \omega(j) = 2 \circ j$, whose cycle type $\lambda \in P(m)$ defines $S(\lambda, F)$. For columns j, j' let $\delta_{j,j'}$ be the row permutation with $\delta(i) \circ j = i \circ j'$; $\{j, j'\}$ is an *A-pair* if rows 1, 2 lie in different cycles of $\delta_{j,j'}$ and a *B-pair* otherwise. The *backflip* at an *A-pair* $\{j, j'\}$ lying on one ω -cycle of length $\alpha + \beta$ with $\omega^\alpha(j) = j'$ trades the $\delta_{j,j'}$ -cycle through row 2; it splits that ω -cycle into the two arcs between j and j' , of lengths α and β (Lemma 3.5). The *cross-switch* at a *B-pair* $\{j, j'\}$ with $\omega^\alpha(j) = j'$ (Definition 3.9) is a Latin bitrade (Q, Q') that preserves $S(\lambda, F)$ (Lemma 3.10).

Let μ be λ with one part $\alpha + \beta$ split into $\alpha, \beta \geq 2$. The splitting procedure (p. 294) considers, for $L \in S(\lambda, F)$, every pair $\{j, j'\}$ with $\omega^\alpha(j) = j'$ on an $(\alpha + \beta)$ -cycle and produces $L' \in S(\mu, F)$ as follows:

1. if $\{j, j'\}$ is an *A-pair*, backflip at $\{j, j'\}$;
2. else if $\{\omega j, \omega j'\}$ is an *A-pair*, backflip at $\{\omega j, \omega j'\}$;
3. otherwise cross-switch with respect to $\{\omega j, \omega j'\}$; “now $\{j, j'\}$ has become an *A-pair*, and we backflip with respect to $\{j, j'\}$ ”.

Item 4 asserts that in each case $L' \in S(\mu, F)$. Together with a fifth edge (a switch) this gives every $L \in S(\lambda, F)$ degree exactly $2(\mu_{\alpha+\beta} + 1)(\alpha + \beta)$ in a bipartite multigraph between $S(\lambda, F)$ and $S(\mu, F)$, while the joining procedure gives every $L \in S(\mu, F)$ degree between $\mu_\alpha \mu_\beta \alpha \beta$ and three times that; the two multigraphs are asserted to coincide, and the double count is Lemma 3.12:

$$\frac{1}{2} \leq \frac{|S(\lambda, F)| (\mu_{\alpha+\beta} + 1)(\alpha + \beta)}{|S(\mu, F)| \mu_\alpha \mu_\beta \alpha \beta} \leq \frac{3}{2} \quad (\alpha \neq \beta). \quad (1)$$

(For $\alpha = \beta$ the statement has $2(\mu_{2\alpha} + 1)/(\alpha \mu_\alpha (\mu_\alpha - 1))$ in place of the fraction; everything below applies verbatim, with the splits (α, α) , $\alpha \geq 3$, in the class $\min(\alpha, \beta) \geq 3$ and $(2, 2)$ the only unaffected split.)

2 The cross-switch reverses an arc

Write $e_1 = 1 \circ j$, $e_2 = 1 \circ j'$, $e_3 = 2 \circ j$, $e_4 = 2 \circ j'$ and consider the first form of Definition 3.9 ($\min(e_1, e_2, e_3, e_4) \in \{e_1, e_4\}$); the second form ($\min \in \{e_2, e_3\}$) is the first form applied to the ordered pair (j', j) , i.e. with the roles of j and j' (and of α and β) exchanged, so that the cross-switch at an unordered B -pair is well defined. On rows 1, 2 the bitrade is

$$Q' \supseteq \{(2, j, e_2), (1, j', e_3)\} \cup \{(1, \omega^k j, 2 \circ \omega^k j), (2, \omega^k j, 1 \circ \omega^k j) : 1 \leq k < \alpha\},$$

i.e. in every interior column of the ω -arc from j to j' the entries of rows 1, 2 are swapped, row 2 receives $e_2 = 1 \circ j'$ at column j , and row 1 receives $e_3 = 2 \circ j$ at column j' . (This is the definition; the proof of Lemma 3.10 confirms it: row 1 keeps the symbol set $\{1 \circ \omega^k j : 1 \leq k \leq \alpha\}$ on the cells $\{\omega^k j : 1 \leq k < \alpha\} \cup \{j'\}$, and “precisely one column from the pair $\{\omega^k j, \omega^{k'} j'\}$ has symbols swapped in rows 1 and 2”.) The remaining cells of Q lie in columns j, j' and in rows other than 1, 2; they do not affect ω .

Lemma 1. *Let L' be the cross-switch of L at the B -pair $\{j, j'\}$, $\omega^\alpha(j) = j'$, and let $\omega' = \omega_{L'}$. Then $\omega'(j) = \omega^{\alpha-1}(j)$, $\omega'(\omega^k j) = \omega^{k-1}(j)$ for $2 \leq k < \alpha$, $\omega'(\omega j) = j'$, and $\omega' = \omega$ on every other column. In words: the interior of the ω -arc from j to j' is traversed in reverse, and the cycle structure of ω is otherwise unchanged.*

Proof. Write $1', 2'$ for the new rows. Then $1' \circ j = e_1$, $1' \circ \omega^k j = 1 \circ \omega^{k+1} j$ for $1 \leq k < \alpha$ (since $2 \circ \omega^k j = 1 \circ \omega^{k+1} j$), $1' \circ j' = 1 \circ \omega j$; $2' \circ j = 1 \circ j' = 1 \circ \omega^\alpha j$, $2' \circ \omega^k j = 1 \circ \omega^k j$ for $1 \leq k < \alpha$, $2' \circ j' = e_4$. Now $\omega'(c)$ is the column whose row-1' entry equals $2' \circ c$: for $c = j$ this is $\omega^{\alpha-1} j$; for $c = \omega^k j$ with $2 \leq k < \alpha$ it is $\omega^{k-1} j$; for $c = \omega j$ it is j' (the symbol $1 \circ \omega j$ now sits at $(1', j')$); for $c = j'$ it is $\omega j'$, unchanged; for $c = \omega^{-1} j$, $2' \circ c = 2 \circ c = e_1 = 1' \circ j$, so $\omega'(c) = j$; all other columns are untouched in both rows. $\square$

The example on p. 293 of [1] shows exactly this: there $\omega = (1\,2\,3\,4\,5\,6\,7)$, $j = 2$, $j' = 5$, $\alpha = 3$, and the displayed cross-switch has rows $1\,2\,4\,5\,3\,6\,7$ and $2\,5\,3\,4\,6\,7\,1$, i.e. $\omega' = (1\,2\,4\,3\,5\,6\,7)$.

3 Case 3 lands in the wrong type

Proposition 2. *Let $\{j, j'\}$ and $\{J, J'\} = \{\omega j, \omega j'\}$ both be B -pairs with $\omega^\alpha j = j'$ on an $(\alpha + \beta)$ -cycle, and let L' be the cross-switch of L at $\{J, J'\}$ (either form). Then $\{j, j'\}$ is an A -pair of L' , but in the first form it is at ω' -distance 2 (from j to j'), and in the second form at ω' -distance 2 from j' to j . Consequently the backflip at $\{j, j'\}$ splits the $(\alpha + \beta)$ -cycle into parts 2 and $\alpha + \beta - 2$, which is the type μ only if $\min(\alpha, \beta) = 2$.*

Proof. First form. The arc from $J = \omega j$ to $J' = \omega^\alpha J$ has interior $\omega J, \dots, \omega^{\alpha-1} J$, and $\omega^{\alpha-1} J = \omega^\alpha j = j'$. By Lemma 1, $\omega'(j) = \omega'(\omega^{-1} J) = J$ and $\omega'(J) = \omega^{\alpha-1} J = j'$. The status of $\{j, j'\}$: column j' is an interior column of the arc, so rows 1, 2 are swapped there, while column j is untouched ($j \notin \{J, J'\}$ and j is not in the arc); hence $\delta_{j, j'}$ changes by a transposition of rows 1, 2 and $\{j, j'\}$ toggles from B to A , as claimed in [1]. The second form reverses the arc from J' to J , whose interior contains $j = \omega^{-1} J$ as its last column, and the same computation gives $\omega'(J') = j$, i.e. $\omega'^2(j') = j$. $\square$

So the asserted membership $L' \in S(\mu, F)$ in item 4 fails in case 3 whenever $\min(\alpha, \beta) \geq 3$.

The case $\min(\alpha, \beta) = 2$. Say $\alpha = 2 \leq \beta$. In the form of the cross-switch that reverses the arc of length 2 (a single interior column), nothing is reversed and case 3 is as described. In the other form the arc of length β is reversed; j or j' is its last interior column, the backflipped square has parts 2 and β , so it does lie in $S(\mu, F)$, and item 4 holds. But the proof of Lemma 3.12 rests on the identity $G_S = G_J$, and this fails: the case-3 edge $L-L'$ is supposed to be the joining rule-4 edge from L' (flip at $\{j, j'\}$ to L'' , then cross-switch at $\{\omega_{L''} j, \omega_{L''} j'\}$), which returns L by Corollary 3.11 only if $\{\omega_{L''} j, \omega_{L''} j'\} = \{J, J'\}$; and when the long arc has been reversed this is not so (e.g. in the second form $\omega_{L''}(j) = \omega^{\beta-2} J' \neq J$). Which form occurs is decided by the position of the smallest corner symbol, not by the author, so about half of the case-3 edges and their switch edges (item 5) are missing from G_J . We computed both multigraphs exactly for $n = 5$, $\lambda = (5)$, $\mu = (2, 3)$, $F = \emptyset$, over all 161280 Latin squares of order 5 (the procedures select trades by the smallest symbol, so one cannot normalise the first row): G_S and G_J each have 1036800 edges, but 172800 edges of G_S are not edges of G_J (exactly half of the case-3 edges and the switch edges of the same instances; all edges of cases 1 and 2 are in G_J), and the degrees of the μ -squares in G_S are 12, 18 and 24 (for 11520, 34560 and 11520 squares) against 18 for every μ -square in G_J . So the per-vertex assertion “each vertex of $S(\mu, F)$ has degree $d \leq 3\mu_\alpha\mu_\beta\alpha\beta$ ” is false for the splitting multigraph, though it holds on average over $S(\mu, F)$; the inequality $Z \leq 3\mu_\alpha\mu_\beta\alpha\beta|S(\mu, F)|$ itself therefore still holds at $n = 5$, with equality (the ratio is exactly 3/2 there), but not for the reason given. For $(4) \rightarrow (2, 2)$ at $n = 4$ the two multigraphs coincide, as they must: only arcs of length 2 occur. In summary, the published argument for the upper bound is incomplete for every split with $\max(\alpha, \beta) \geq 3$.

Numerical confirmation. We implemented Definition 3.9 literally (rows 1, 2 as above; in columns j, j' the rows $\delta^k(1)$, $1 \leq k < a$, with $\delta^a(1) = 2$, swap their two entries, exactly as in the definition — this is what balances the corner cells) and checked on uniformly sampled Latin squares (Jacobson–Matthews) that the cross-switch is a Latin square of the same type in every instance. Applying case 3 as written: in 1006 ordered instances (j, j') with $\min(\alpha, \beta) \geq 3$ at $n = 9$ (300 sampled squares) and 3098 at $n = 12$ (300 squares), the result had the wrong type in every single instance (e.g. $(2, 7)$ with $(\alpha, \beta) = (3, 4)$ gives $(2, 2, 5)$ instead of $(2, 3, 4)$; (9) with $(3, 6)$ gives $(2, 7)$).

4 What survives, and what depends on the gap

Only case 3 is affected. The joining rules 1–2 are valid and each rule-2 edge is a case-1 edge or the switch edge of a case-1 instance, so each $L \in S(\mu, F)$ has degree at least $\mu_\alpha\mu_\beta\alpha\beta$ in the multigraph of valid splitting edges, in which each $L \in S(\lambda, F)$ has degree at most $2(\mu_{\alpha+\beta} +$

1)($\alpha + \beta$). Hence the *lower* bound $\frac{1}{2}$ in (1) stands. (In fact it is elementary: with X the set of pairs $(L, \{j, j'\})$, $L \in S(\lambda, F)$, $\omega^\alpha j = j'$, and J the analogous set of joining pairs, the flip and the switch give $|J| = 2|X_A|$ exactly, so (1) reads $\frac{1}{3} \leq |X_A|/|X| \leq 1$, and the lower bound $\frac{1}{2}$ is $|X| \geq |X_A|$.) The *upper* bound $\frac{3}{2}$, equivalently

$$\Pr[\{j, j'\} \text{ is an } A\text{-pair}] \geq \frac{1}{3} \quad \text{for } (L, \{j, j'\}) \text{ uniform on } X,$$

is not established for $\max(\alpha, \beta) \geq 3$.

Within [1] the upper bound of Lemma 3.12 is used for: the upper bound of Theorem 3.13; the lower bound $(2/3)^{\kappa-1}$ of Lemma 4.1 and, in Lemma 4.2, the upper bound whenever the path from λ to μ contains a splitting step and the lower bound whenever it contains a joining step; the bounds on $\mathbb{P}_F(E)$ of Theorem 4.3 (whose proof invokes the equality of the two multigraphs “as in the proof of Lemma 3.12”); the upper bound $\mathbb{P}_F((m)) \leq 2m^{-2/3}$ of Lemma 4.4; both halves of Corollary 4.5 (its lower half uses the $(2/3)^{\kappa-1}$ bound, its upper half the factor $m^{1/3}$ from Lemma 4.4); and the results that rest on these: Corollary 4.6, Lemma 4.7, Theorem 4.9, Corollary 4.10 and Theorem 4.11. (Corollary 4.10, on the number of intercalates, is in any case superseded by Kwan and Sudakov [2].) The upper bound $2^{\kappa-1}$ of Lemma 4.1, the lower bound $\mathbb{P}_F((m)) \geq 2m^{-2}$ of Lemma 4.4, Corollary 4.8, and the upper bounds on short-cycle counts derived from the joining direction are unaffected. For instance, from the joining direction alone one still gets $\mathbb{E}[\ell N_\ell] \leq 2m/(m - \ell)$ for the number N_ℓ of ℓ -cycles, $\ell \leq m/2$ (sum the inequality $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta)N_{\ell+\beta}]$, with $N_\ell N_\beta$ read as $N_\ell(N_\ell - 1)$ when $\beta = \ell$, over β), and by joining k cycles successively one obtains a factorial-moment bound of the form $\mathbb{E}[(\kappa)_k] \leq 2^{k-1}(4 + k^2/\log m)(\log m)^k$ for the number of cycles κ , hence $\Pr[\kappa \geq 16 \log m] = O(m^{-2} \log m)$ (we give the derivation elsewhere). So a polynomial tail bound on the number of cycles at scale $\log m$ survives; the bound of Theorem 4.9 at scale $\sqrt{m}$, and the lower bound on the number of cycles of Theorem 4.11, do not.

5 A partial repair

A cross-switch at a B -pair $\{J, J'\}$ at any ω -distance $a \geq 2$ is a type-preserving trade (the proof of Lemma 3.10 does not use $a = \alpha$), and by Lemma 1 it reverses the arc from J to J' . If $\{J, J'\}$ is disjoint from $\{j, j'\}$, it toggles the status of $\{j, j'\}$ iff exactly one of j, j' is an interior column of that arc, say at position i ($1 \leq i < a$), and then the ω -distance from j to j' changes by $\pm(a - 2i)$ (sign according to which of j, j' is interior). Thus, among cross-switches at pairs disjoint from $\{j, j'\}$, the status-toggling moves that preserve the distance are precisely those at B -pairs *symmetric about j or about j'* ($a = 2i$).

Suppose α is even and $\alpha < 2\beta$, and let $R = \{\omega^{\alpha/2}j, \omega^{3\alpha/2}j\}$: the pair at distance α symmetric about j' , whose arc avoids j . If R is a B -pair, cross-switch at R : $\{j, j'\}$ becomes an A -pair whose two ω' -arcs still have lengths α and β (in the form that reverses the arc of R through j' the distance from j to j' stays α ; in the other form it becomes β), and the backflip at $\{j, j'\}$ lies in $S(\mu, F)$. If R is an A -pair, backflip at R itself, which is a pair at distance α . Either way case 3 acquires a valid edge (verified on 3928 sampled instances at $n = 10, 13$: type μ in all of them). The same works with the roles of α, β exchanged when β is even and $\beta < 2\alpha$. The multiplicities on the $S(\mu, F)$ side change (an A -pair’s backflip may now serve two splitting pairs), so the constant $3/2$ would have to be recomputed.

When α and β are both odd, every distance-preserving toggle has even arc length, so no symmetric pair is at distance α or β ; if all symmetric pairs about j and j' happen to be A -pairs,

no single cross-switch repairs $\{j, j'\}$, and two-step repairs pass through a third type, which breaks the two-sided count. We do not see a canonical repair in this case (nor when $\alpha \geq 2\beta$ with β odd). Repairs that use other rows do exist — for a B -pair $\{j, j'\}$ with column cycle $Z \ni 1, 2$, trade rows z, w lying on different arcs of Z between 1 and 2 along the row cycle of $\{z, w\}$ through j (if j, j' lie in different row cycles of $\{z, w\}$; this separates 1 from 2 and keeps the type), or cross-switch rows z, w at $\{j, j'\}$ (if they lie in one; this changes the type to μ) — but they are indexed by a free pair of rows and so have unbounded multiplicity on the $S(\mu, F)$ side, which defeats the double count.

6 What the data say

The statement of Lemma 3.12 is almost certainly true, and far from sharp for $n \geq 6$. By the identity $|J| = 2|X_A|$ the quantity in (1) is $\mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$, where $\mathcal{C}_n(\lambda)$ is the number of completions of two rows of type λ , and $\Pr_X[A] = \mathcal{C}_n(\mu)/(2\mathcal{C}_n(\lambda))$. From the exact values of $\mathcal{C}_n(\lambda)$ tabulated in [1] (Tables 1 and 2, $n \leq 11$):

n	#splits $(\lambda \rightarrow \mu)$	$\max \mathcal{C}_n(\lambda)/\mathcal{C}_n(\mu)$	$\min \Pr_X[A]$
5	1	1.5000 (sharp)	0.3333
6	3	0.9545	0.5238
7	4	1.0035	0.4982
8	8	1.0017	0.4991
9	11	1.0002	0.4999
10	19	1.0000	0.5000
11	26	1.0000	0.5000

(all splits with $\alpha, \beta \geq 2$; for $n \geq 7$ every ratio lies in $[0.985, 1.0035]$, with the minima 0.9861 at $(3, 4) \rightarrow (2, 2, 3)$, $n = 7$, and 0.9853 at $(2, 2, 4) \rightarrow (2, 2, 2, 2)$, $n = 8$, and the excess above 1 at most 0.35%, in the covered and the uncovered classes alike — e.g. $(2, 6) \rightarrow (2, 3, 3)$ at $n = 8$, in the $(3, 3)$ class, has ratio 1.0017). On uniformly sampled squares with a uniformly random marked pair at distance $\alpha \geq 2$ on an m -cycle, $m - \alpha \geq 2$, we find $\Pr_X[B] - \Pr_X[A] = -0.022 \pm 0.011, +0.008 \pm 0.012, -0.030 \pm 0.018$ at $n = 30, 50, 100$ (9000, 7200, 3600 instances), i.e. $\Pr_X[A] = 0.50 \pm 0.01$ throughout. So the true constant in the upper bound of (1) appears to be $1 + o(1)$ (the maximal excess over 1 in the table is $3.5 \cdot 10^{-3}, 1.7 \cdot 10^{-3}, 1.8 \cdot 10^{-4}, < 10^{-5}, < 10^{-5}$ for $n = 7, \dots, 11$) rather than $3/2$; the question is only what has been proved.

References

- [1] N. J. Cavenagh, C. Greenhill and I. M. Wanless, The cycle structure of two rows in a random Latin square, *Random Structures & Algorithms* 33 (2008), 286–309.
- [2] M. Kwan and B. Sudakov, Intercalates and discrepancy in random Latin squares, *Random Structures & Algorithms* 52 (2018), 181–196.